Power Series and Taylor Series

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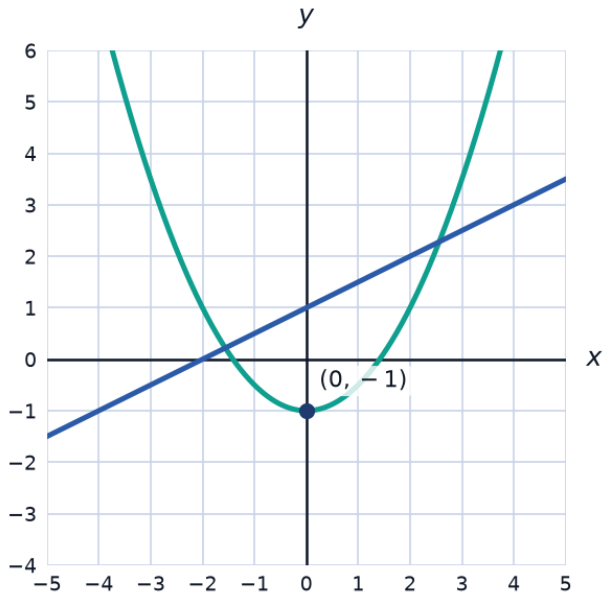
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Radius and interval of convergence

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$.

2 [Foundational] • Target: Radius and interval of convergence

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.



3 [Foundational] • Target: Taylor and Maclaurin series

Find the first three nonzero terms of the Maclaurin series for $f(x) = e^x$, using $f^{(n)}(0) = 1$ for all n.

4 [Foundational] • Target: Taylor and Maclaurin series

Find the third-degree Taylor polynomial for $f(x) = \cos x$ centered at $x = 0$.

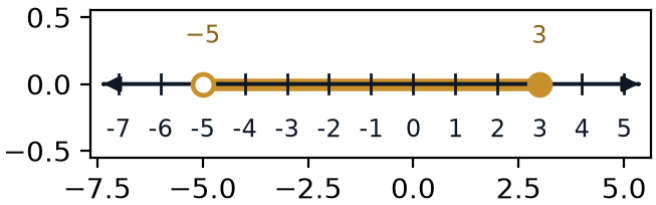
5 [Foundational] • Target: Taylor and Maclaurin series

Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$ to approximate $\sin(0.2)$ using the first two nonzero terms, to four decimal places.

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6 [Foundational] • Target: Interval of convergence (including endpoints)

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n}$, checking both endpoints.



7 [Foundational] • Target: Interval of convergence (including endpoints)

Find the radius of convergence of $\sum_{n=1}^{\infty} n!\, x^n$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Using known series

Using the known Maclaurin series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, write the Maclaurin series for e^{-x^2} (substitute $-x^2$ for x), giving the first three terms.

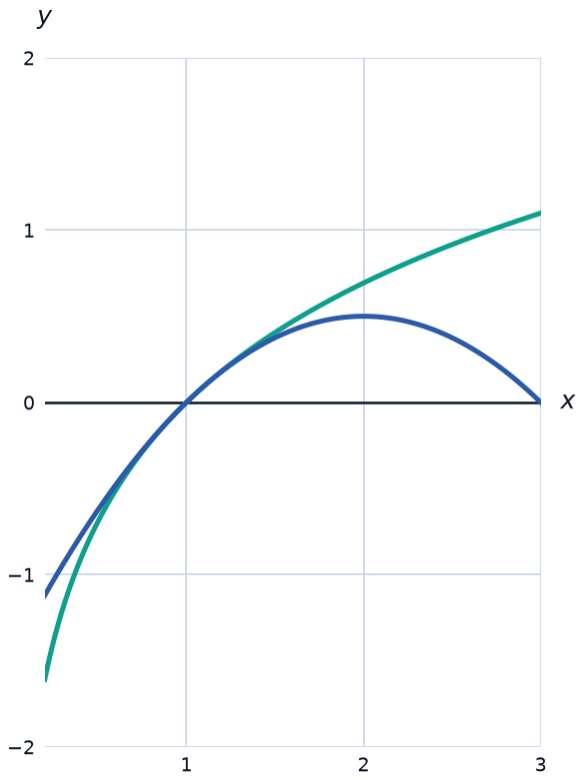
9 [Intermediate] • Target: Using known series

Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, write the power series for $\frac{1}{1+x^2}$ (substitute $-x^2$ for x).

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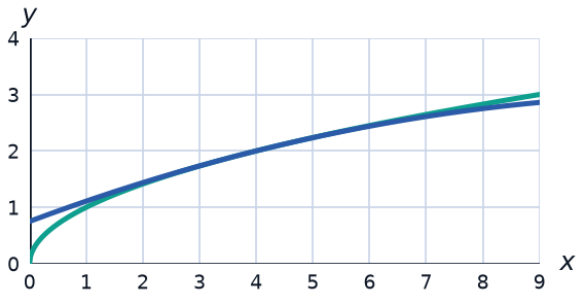
10 **[Intermediate]** • **Target: Taylor series centered at a nonzero point**

Find the first three terms of the Taylor series for $f(x) = \ln x$ centered at $x = 1$, using $f(1) = 0$, $f'(1) = 1$, $f''(1) = -1$.



11 **[Intermediate]** • **Target: Taylor series centered at a nonzero point**

Find the second-degree Taylor polynomial for $f(x) = \sqrt{x}$ centered at $x = 4$, given $f(4) = 2$, $f'(4) = \frac{1}{4}$, $f''(4) = -\frac{1}{32}$.



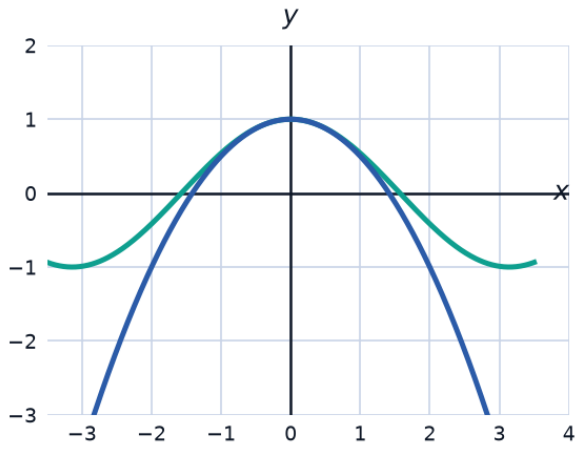
12 **[Intermediate]** • **Target: Error estimation with Taylor polynomials**

Using the second-degree Maclaurin polynomial for $\cos x$, $T_2(x) = 1 - \frac{x^2}{2}$, estimate $\cos(0.3)$ to four decimal places, and compare to the true value $\cos(0.3) \approx 0.9553$.

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13 [Intermediate] • Target: Visualizing a Taylor approximation

The graph compares $f(x) = \cos x$ to its second-degree Maclaurin polynomial $T_2(x) = 1 - \frac{x^2}{2}$. Use the graph to describe where the approximation is best and where it breaks down.



14 [Intermediate] • Target: Differentiating and integrating power series

Given $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, differentiate both sides term-by-term to find a power series for $\frac{1}{(1-x)^2}$.

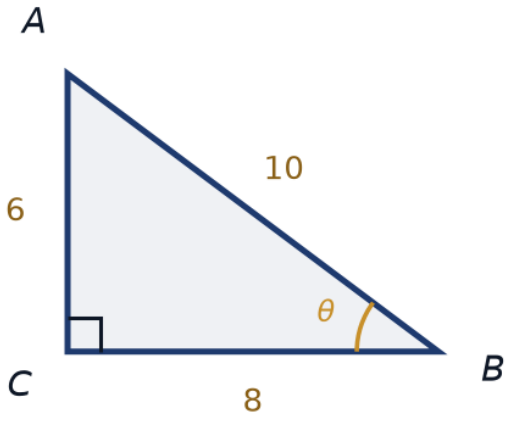
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Differentiating and integrating power series • Cross-Lesson: 7.02 Integration by Parts

Using the Maclaurin series $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$, integrate term-by-term to find the first three terms of the power series for $\int \sin x \, dx = -\cos x + C$, and confirm they match the known Maclaurin series for $-\cos x$.

16 [Challenging] • Target: Using Taylor series to evaluate a limit • Cross-Lesson: 8.01 Area Between Curves

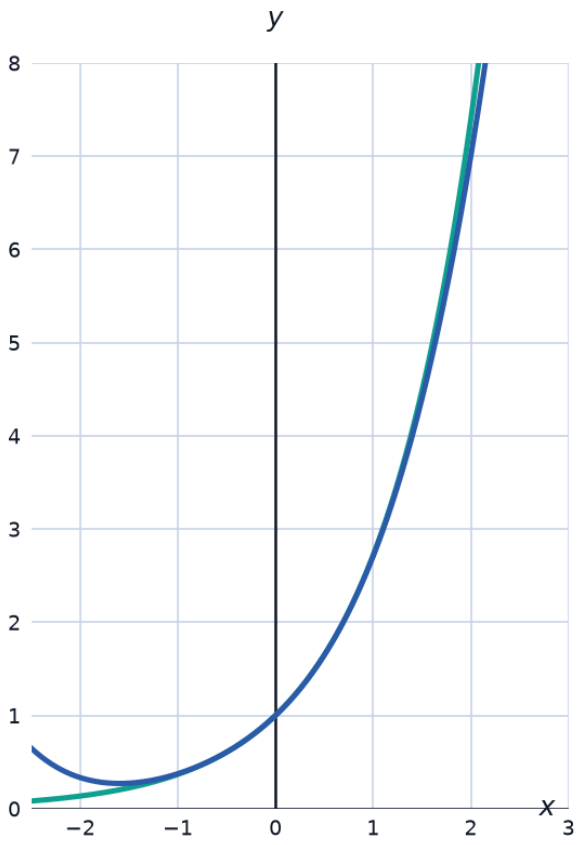
Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \cdots$ to evaluate $\lim_{x \to 0} \frac{\sin x - x}{x^3}$.



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17 [Challenging] • Target: Using Taylor series to evaluate a limit • Cross-Lesson: 8.02 Volumes of Revolution

Find the fourth-degree Taylor polynomial for $f(x) = e^x$ centered at $x = 0$, and use it to estimate $e^{0.5}$, comparing to the true value $e^{0.5} \approx 1.6487$.



18 [Challenging] • Target: Using Taylor series to evaluate a limit • Cross-Lesson: 8.03 Arc Length and Improper Integrals

An engineer needs a quick approximation for $\ln(1.1)$ without a calculator. Using the Maclaurin series $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ for $|x| < 1$, find the approximation using the first three terms with $x = 0.1$, and compare to the true value $\ln(1.1) \approx 0.09531$.

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19 [Challenging] • Target: Error estimation with Taylor polynomials • Cross-Lesson: 9.01 Separable Differential Equations

Using Stokes' Theorem, evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = \langle -y^2, x, z^2 \rangle$ and C is the intersection of $z = y$ and the cylinder $x^2 + y^2 = 1$ (oriented counterclockwise from above).



20 [Challenging] • Target: Visualizing a Taylor approximation • Cross-Lesson: 1.01 Precalculus Review for Calculus

Find the coefficient of x^3 in the Maclaurin series of $f(x) = \frac{\sin(x)}{1-x}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Radius and Interval of Convergence

Find the radius of convergence R for the power series $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n \cdot 3^n}$.

a. $R = 3$

b. $R = 1/3$

c. $R = 1$

d. $R = \infty$

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22 [Foundational] • Target: Radius and Interval of Convergence

Find the exact interval of convergence for $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n \cdot 3^n}$.

a. $[-1, 5)$

b. $(-1, 5)$

c. $[-1, 5]$

d. $(-1, 5]$

23 [Foundational] • Target: Radius and Interval of Convergence

Find the radius of convergence of $\sum_{n=0}^{\infty} n!\,(2x)^n$.

a. $R = 0$

b. $R = 1/2$

c. $R = 1$

d. $R = \infty$

24 [Foundational] • Target: Geometric Series Representations

Find the power series representation for $f(x) = \frac{1}{1+2x}$ centered at 0.

a. $\sum_{n=0}^{\infty} (-1)^n 2^n x^n$

b. $\sum_{n=0}^{\infty} 2^n x^n$

c. $\sum_{n=0}^{\infty} (-1)^n x^n$

d. $\sum_{n=1}^{\infty} (-2x)^n$

25 [Foundational] • Target: Geometric Series Representations

Find the Maclaurin series for $\frac{x}{4-x^2}$.

a. $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{4^{n+1}}$

b. $\sum_{n=0}^{\infty} \frac{x^{2n}}{4^n}$

c. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{4^{n+1}}$

d. $\frac{1}{4}\sum_{n=0}^{\infty} x^{2n+1}$

26 [Foundational] • Target: Geometric Series Representations

Find the Maclaurin series for $\ln(1+x)$ by integrating the series for $\frac{1}{1+x}$.

a. $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$

b. $\sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$

c. Both of the above are equivalent

d. $\sum_{n=1}^{\infty} \frac{x^n}{n}$

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27 [Foundational] • Target: Standard Maclaurin Series

State the Maclaurin series for e^x.

a. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

b. $\sum_{n=1}^{\infty} \frac{x^n}{n}$

c. $\sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n!}$

d. $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

28 [Foundational] • Target: Standard Maclaurin Series

State the Maclaurin series for $\cos(x)$.

a. $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$

b. $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$

c. $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

d. $\sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n-1)!}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Standard Maclaurin Series

Find the first three non-zero terms of the Maclaurin series for $\sin(2x)$.

a. $2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$

b. $2x - 4x^3 + \frac{4}{3}x^5$

c. $2x - \frac{8}{6}x^3 + \frac{32}{120}x^5 = 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$

d. Both of the above are correct

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30 [Intermediate] • Target: Taylor Polynomials and Lagrange Error Bound

Find the 2nd-degree Taylor polynomial $T_2(x)$ for $f(x) = \sqrt{x}$ centered at $a = 4$.

a. $T_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$

b. $T_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{32}(x-4)^2$

c. $T_2(x) = 2 + \frac{1}{2}(x-4) - \frac{1}{16}(x-4)^2$

d. $T_2(x) = 2 + \frac{1}{4}(x-4) + \frac{1}{64}(x-4)^2$

31 [Intermediate] • Target: Taylor Polynomials and Lagrange Error Bound

State the formula for the Lagrange Error Bound for a Taylor polynomial $T_n(x)$ centered at a.

a. $|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$, where $M = \max|f^{(n+1)}(t)|$ on $[a, x]$

b. $|R_n(x)| \le \frac{M}{n!}|x-a|^n$

c. $|R_n(x)| \le M|x-a|^{n+1}$

d. $|R_n(x)| \le \frac{f^{(n+1)}(a)}{(n+1)!}|x-a|^{n+1}$

32 [Intermediate] • Target: Taylor Polynomials and Lagrange Error Bound

Use the 3rd-degree Maclaurin polynomial for e^x to bound the error in estimating $e^{0.1}$. (Take $e^{0.1} < 3$).

a. $|R_3(0.1)| \le \frac{3}{24}(0.1)^4 = 1.25 \times 10^{-5}$

b. $|R_3(0.1)| \le 10^{-4}$

c. $|R_3(0.1)| \le \frac{1}{6}(0.1)^3$

d. $|R_3(0.1)| \le 10^{-6}$

33 [Intermediate] • Target: Taylor Polynomials and Lagrange Error Bound

Interval of convergence of $\sum \frac{(-3)^n(x-2)^n}{\sqrt{n}}$: centre, radius, and endpoints:

a. $R = 1/3$, interval $(5/3, 7/3]$

b. $R = 1/3$, interval $[5/3, 7/3)$

c. $R = 1/3$, interval $[5/3, 7/3]$

d. $R = 3$, interval $(-1, 5]$

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34 [Intermediate] • Target: Radius and interval of convergence

Consider the power series $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2 3^n}$. This series defines a function $f(x)$ over its interval of convergence. Let I be the interval of convergence for this power series. Determine which of the following statements is TRUE about $f(x)$ and its behavior at the endpoints of I.

a. The radius of convergence is 3, and the series converges at both endpoints of the interval of convergence.

b. The radius of convergence is 3, and the series diverges at the left endpoint and converges at the right endpoint.

c. The radius of convergence is $1/3$, and the series converges at both endpoints of the interval of convergence.

d. The radius of convergence is $1/3$, and the series diverges at both endpoints of the interval of convergence.

35 [Intermediate] • Target: Taylor and Maclaurin series

The error function $\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$. Using $e^{-t^2} = \sum (-1)^n t^{2n}/n!$:

a. $\text{erf}(x) = \sum x^n/n!$

b. $\text{erf}(x) = \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n!\,(2n+1)}$

c. $\text{erf}(x) = e^{-x^2}$

d. $\text{erf}(x) = 2x/\sqrt{\pi}$

36 [Intermediate] • Target: Interval of convergence (including endpoints)

Error bound: $|\cos x - 1 + x^2/2| \le x^4/24$ (from T_4 bound). This means the error at $x = 0.1$ is at most:

a. $0.0001/24$

b. $0.1^4/24$

c. $0.1^2/2$

d. $1/24$

37 [Intermediate] • Target: Using known series

The interval of convergence always contains at least the point:

a. $x = R$

b. $x = 1$

c. $x = -1$

d. $x = 0$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Taylor series centered at a nonzero point** • **Cross-Lesson: 8.03 Arc Length and Improper Integrals**

The error in using $T_n(x)$ decreases as n increases (assuming convergence). This is because the remainder $R_n \to$

a. 1

b. $f(x)$

c. $n!$

d. 0

39 **[Challenging]** • **Target: Error estimation with Taylor polynomials** • **Cross-Lesson: 9.01 Separable Differential Equations**

The integral $\int_0^1 e^{-x^2} dx$ (error function) — the exact value cannot be expressed in elementary functions. Using the series, approximate as $\sum \frac{(-1)^n}{n!\,(2n+1)}$. First two terms:

a. $1 - 1/6 \approx 0.833$

b. $1 - 1/3 \approx 0.667$

c. $1 + 1/3 \approx 1.333$

d. 0

40 **[Challenging]** • **Target: Visualizing a Taylor approximation** • **Cross-Lesson: 1.01 Precalculus Review for Calculus**

Maclaurin series for $\tan^{-1} x$ at $x = 1$ gives $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \cdots$. This converges very slowly. The 10th partial sum approximates $\pi/4$ with error less than:

a. 1/11

b. 1/21

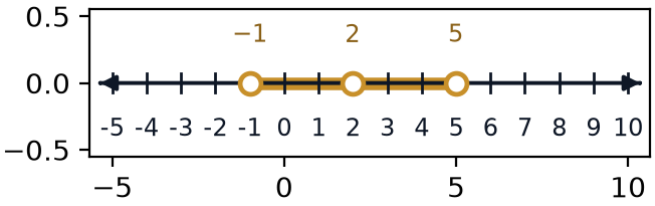
c. 1/19

d. 0.01

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41 [Challenging] • Target: Differentiating and integrating power series • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Consider the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$. If the radius of convergence is $R=3$ and the center of the series is $a=2$, which of the following intervals represents the values of x for which the series converges?



a. $(-1,5)$

b. $[-1,5]$

c. $(-1,5]$

d. $[-1,5)$

42 [Challenging] • Target: Using Taylor series to evaluate a limit • Cross-Lesson: 11.01 Sequences and Series

Find the interval of convergence of $\sum\frac{(x-1)^n}{n3^n}$:

a. $[-2,4)$

b. $(-2,4)$

c. $(-2,4]$

d. $[-2,4]$

43 [Challenging] • Target: Radius and Interval of Convergence • Cross-Lesson: 12.01 Convergence Tests

Find the interval of convergence of $\sum_{n=1}^{\infty}\frac{5^{-n}(x+3)^n}{n^2}$.

a. $[-8,2]$

b. $(-8,2)$

c. $[-8,2)$

d. $(-8,2]$

44 [Challenging] • Target: Geometric Series Representations • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Find the interval of convergence of $\sum_{n=1}^{\infty}\frac{5^{-n}(x+2)^n}{n^2}$.

a. $[-7,3]$

b. $(-7,3)$

c. $[-7,3)$

d. $(-7,3]$

45 [Challenging] • Target: Standard Maclaurin Series • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Find the interval of convergence of $\sum_{n=1}^{\infty}\frac{3^{-n}(x+1)^n}{n^2}$.

a. $[-4,2]$

b. $(-4,2)$

c. $[-4,2)$

d. $(-4,2]$

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